

Qatar-2b

R-0951

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170521 · created 2026-07-29T14:38:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457894.69182 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.067
Transit depth (Rp/Rs)^2	3.04 +/- 2.33 [%]
Orbital Inclination (inc)	84.0 +/- 1.74
Airmass coefficient 1 (a1)	1.707 +/- 0.039
Airmass coefficient 2 (a2)	-0.38 +/- 1.34
Scatter in the residuals of the lightcurve fit is	2.28 %
Best Comparison Star	#3 - [607, 202]
Optimal Aperture	8.66
Optimal Annulus	21.05
Transit Duration (day)	0.049 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.067 vs 0.16327 (deviation 0.16 σ)
Duration fit vs published	0.049 d vs 0.0767 d (deviation 3.79 σ)
Mid-time O-C	1.2 min
Depth SNR	2.6
Residual scatter	2.28 %

Light curve

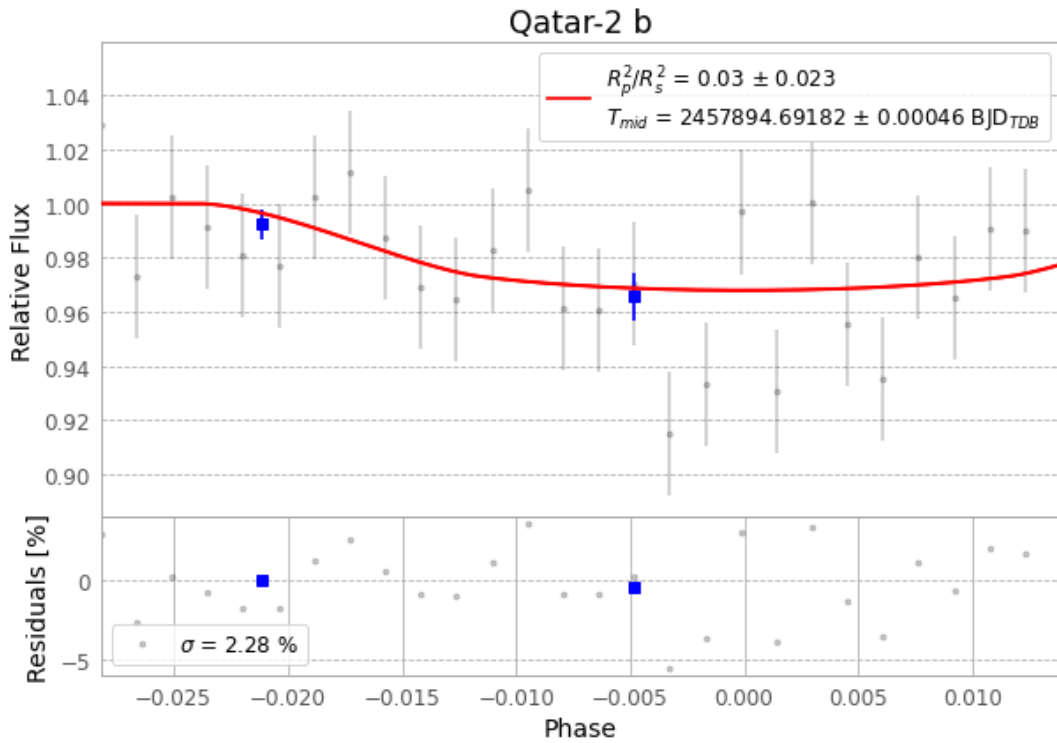


Figure 1 — FinalLightCurve_Qatar-2 b_2017-05-20.png (SHA-256 93a9fe392f6fed92...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [448, 424], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5f94e120e61166f8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

64 FITS frames (42 MB), Qatar-2170521033312.FITS → Qatar-2170521064213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170521.log (3891df09099c...), FinalLightCurve_Qatar-2 b_2017-05-20.png (93a9fe392f6f...), FinalLightCurve_Qatar-2 b_2017-05-20.csv (63d0f5b5f221...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:38Z.